WORKED SOLUTION

STEP II 2001, Question 13

Probability and Statistics • Continuous Distributions and Estimation

STEP II 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q13

The key idea

We are handed the survival function $\mathrm{P}(T > t)$ directly, so the median comes from setting it equal to $\frac{1}{2}$, and the mean from differentiating to get the density and then integrating. The experiment gives a sample median (the fifth of nine ordered lifetimes) and a sample mean, and matching these to the formulas gives two equations in α and k. Dividing one by the other removes k, leaving an equation for α alone. The final probability is a conditional probability, and a pleasant surprise is waiting there.

Solution

The median and the mean

Write $S(t) = \mathrm{P}(T > t) = (1+kt)^{-\alpha}$ for $t \geq 0$. For this to fall from 1 to 0 we need $k > 0$.

Median. The median M satisfies $\mathrm{P}(T > M) = \frac{1}{2}$, so

$$(1+kM)^{\alpha} = 2 \quad \Longrightarrow \quad 1 + kM = 2^{1/\alpha} \quad \Longrightarrow \quad M = \frac{2^{1/\alpha} - 1}{k}.$$

Mean. The cumulative distribution function is $F(t) = 1 - (1+kt)^{-\alpha}$, so the probability density function is

$$f(t) = F'(t) = \alpha k \, (1+kt)^{-\alpha-1}, \qquad t \geq 0.$$

Then, integrating by parts with $u = t$ and $\mathrm{d}v/\mathrm{d}t = \alpha k(1+kt)^{-\alpha-1}$, so that $v = -(1+kt)^{-\alpha}$,

$$\begin{aligned} m = \int_0^{\infty} t\,\alpha k(1+kt)^{-\alpha-1}\,\mathrm{d}t &= \Big[-t(1+kt)^{-\alpha}\Big]_0^{\infty} + \int_0^{\infty} (1+kt)^{-\alpha}\,\mathrm{d}t \\ &= 0 + \left[-\frac{(1+kt)^{1-\alpha}}{k(\alpha-1)}\right]_0^{\infty} = \frac{1}{k(\alpha-1)}. \end{aligned}$$

Both limits at infinity use $\alpha > 1$: $t(1+kt)^{-\alpha}$ behaves like $t^{1-\alpha}/k^{\alpha} \to 0$, and $(1+kt)^{1-\alpha} \to 0$. (If $\alpha \leq 1$ the mean would be infinite, which is why the question insists on $\alpha > 1$.) So

$$M = \frac{2^{1/\alpha} - 1}{k}, \qquad m = \frac{1}{k(\alpha - 1)}.$$

Estimating α and k

With nine bulbs, the fifth failure is the middle one of the nine ordered lifetimes: it is the *sample median*. So we estimate $M \approx 1000$ and, from the sample mean, $m \approx 2400$.

Ask yourself We have two equations and two unknowns, but k appears in both. How can we get at α on its own?

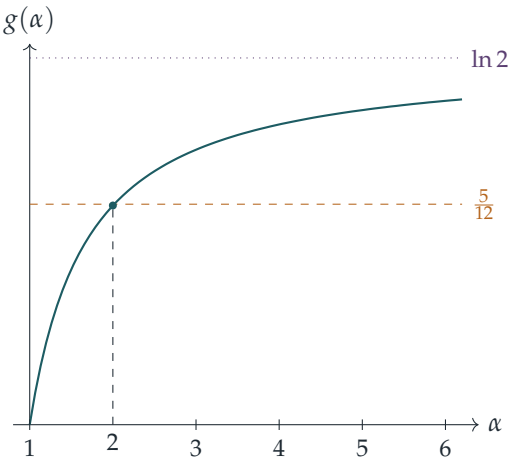
Divide: k cancels, since both M and m are proportional to $1/k$. That gives

$$\frac{M}{m} = (2^{1/\alpha} - 1)(\alpha - 1) \approx \frac{1000}{2400} = \frac{5}{12} \approx 0.4167.$$

Call the left-hand side $g(\alpha)$. At $\alpha = 2$,

$$g(2) = (\sqrt{2} - 1)(2 - 1) = \sqrt{2} - 1 \approx 0.4142,$$

which is within 0.003 of $5/12$. To see that no other value fits as well, note that $g(1.9) \approx 0.396$ and $g(2.1) \approx 0.430$, so the value $5/12$ lies between them. In fact g increases steadily from $g(1) = 0$ towards $\ln 2 \approx 0.693$ as $\alpha \to \infty$ (see the graph), so it takes the value $5/12$ only once, at an α just above 2. Hence $\alpha \approx 2$.



Now put $\alpha = 2$ back into either formula. From the mean,

$$k = \frac{1}{m(\alpha - 1)} \approx \frac{1}{2400},$$

and from the median, $k = (\sqrt{2} - 1)/1000 \approx 4.14 \times 10^{-4}$. The two agree closely, and

$$k \approx 4.2 \times 10^{-4} \text{ per hour.}$$

The conditional probability

A bulb that lasts M hours and then a further $m - M$ hours has lasted m hours in total. Since $m > M$, the event $\{T > m\}$ is contained in $\{T > M\}$, so

$$\mathrm{P}(T > m \mid T > M) = \frac{\mathrm{P}(T > m)}{\mathrm{P}(T > M)} = \frac{(1+km)^{-\alpha}}{1/2} = \frac{2}{(1+km)^{\alpha}}.$$

(Strictly, the bulb is found to be still working at M hours; for a continuous lifetime, $T \geq M$ and $T > M$ have the same probability.) Now the surprise: $km = 1/(\alpha - 1)$ exactly, whatever k is, so $1 + km = \alpha/(\alpha - 1)$ and

$$\mathrm{P}(T > m \mid T > M) = 2\left(\frac{\alpha - 1}{\alpha}\right)^{\alpha}.$$

With $\alpha = 2$ this is $2 \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{2}$. So the estimated probability is

$$\mathrm{P}(\text{lasts a further } m - M \text{ hours}) \approx \tfrac{1}{2}.$$

The estimate of k is not needed at all for this last step.

Checks

- At the median, $S(M) = (2^{1/\alpha})^{-\alpha} = \frac{1}{2}$, as it should be. Numerical integration of $S(t)$ over $[0, \infty)$ for random $\alpha > 1$ and $k > 0$ reproduces $m = 1/(k(\alpha - 1))$ (recall that for a non-negative variable $\mathrm{E}(T) = \int_0^{\infty} \mathrm{P}(T > t)\,\mathrm{d}t$, which is exactly the second integral in our working).
- Solving $g(\alpha) = 5/12$ exactly gives $\alpha \approx 2.015$, and then the conditional probability is $2(1 - 1/\alpha)^{\alpha} \approx 0.502$, so rounding to $\alpha = 2$ costs almost nothing.
- A simulation of 400 000 lifetimes with $\alpha = 2$, $k = 1/2400$ (taking $T = (U^{-1/2} - 1)/k$ for U uniform on $(0, 1)$) gave $\mathrm{P}(T > m \mid T > M) \approx 0.501$.

Common slips

- Treating $\mathrm{P}(T > t)$ as the cumulative distribution function and differentiating it without the minus sign, giving a negative "density".
- Answering $\mathrm{P}(T > m - M)$, as if the bulb started afresh at time M. This distribution is not memoryless (only the exponential is), so the condition $T > M$ matters.
- Using the fifth failure as an estimate of the mean rather than the median.

Beyond STEP

This is a Pareto-type (Lomax) distribution, a standard model for lifetimes and for incomes, with a heavy tail: $\mathrm{E}(T^r)$ is finite only for $r < \alpha$. What we did here is the *method*

of moments (strictly, matching a quantile and a moment): choose parameters so that the model's summary statistics equal the sample's. First-year statistics compares this with maximum likelihood and asks how accurate such estimates are from only nine bulbs. The ratio $\mathrm{P}(T > s + t \mid T > s)$ is the survival function used in reliability theory: here it *increases* with s, so old bulbs are more likely to keep going than new ones. Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.